Stage 3 – Use Case Documentation Template

Section 1 – Team Accomplishments & Task Assignments

A. Summary of Team Accomplishments
Describe what your team completed during this stage of the project:

- Met with instructor and received feedback

- Refined user stories

- Finalized approved user stories

- Began implementation planning

Overall, our user stories were good. We also decided for simplicity to only base this application on the UTRGV cybersecurity degree works and degree map.

Feedback from Dr. Zhang:

- Should include user stories for viewing transcripts, degree plans, and course recommendations for both the faculty and student roles

In Stage 3, we will address this feedback accordingly.

B. Task Assignments Per Team Member

The three of us had the same tasks because we met to talk about the feedback from Dr. Zhang, and we refind our user stories, and then we all three worked together to produce the use cases that reflect the refined user stories. We also designed the alternative flows for each.

Section 2 – Final Approved User Stories
List the user stories approved by your instructor. Use the proper format:

1. As a new user, I want to register with a username and password, so that I can securely access the Advising Assistant System.

2. As a registered user, I want to log into the system, so that I can access advising features based on my role.
3. As a system, I want to redirect access based on user role (faculty or student), so that users can only access authorized data
4. As a faculty member, I want to view a list of my advised students, so that I can select a student to review their academic information.
5. As an advisor, I want to view the transcript of an advised student, so that I can review their completed coursework.
6. As an advisor, I want to generate a report containing recommended future courses and completed coursework so that students receive academic guidance.
7. As an advisor, I want to view a student's major degree plan, so that I can track their academic progress.
8. As an advisor, I want to view prerequisites for any course in degree plan, so that I can prevent students from enrolling in ineligible courses.
9. As a student, I want to view my own transcript, so that I can monitor my academic progress.
10. As a student, I want to view my degree plan, so that I can see remaining required courses.
11. As a student, I want to view recommended courses for next semester, so that I can plan my schedule effectively.
12. As a logged-in user, I want to log out of the system, so that my account remains secure.

As a system, I want course names to function as links that display detailed course descriptions and prerequisites, so that users can easily and quickly access course information.

As a student, I want to be able to see if a course has any prerequisites, so I know if I am eligible to take the course or not.

Section 3 – Use Cases

1. Use Case Name: Register User Account

   Primary Actor: Student or Faculty

   Goal: User is able to register account

   Preconditions:

   - User does not already have an account
   - System is operational

   Postconditions:

   - Account is successfully created
   - User credentials are stored securely

   Basic Flow:

   1. User navigates to Registration page.
   2. User enters username and password.
   3. User selects role (Student or Faculty).
   4. User submits form.
   5. System validates input.
   6. System creates account.
   7. System redirects user to Login page.

Alternative Flow 1 (Username Already Exists):

Occurs at Step #: 5

1. At Step 5, system detects duplicate username.

2. System displays error message.

3. User is prompted to choose a different username.

2. Use Case Name: Login to System

Primary Actor: Registered User

Goal: User is able to login to system

Preconditions:

- User account exists

- System is operational

Postconditions:

- Account is successfully created

- User credentials are stored securely

Basic Flow:

1. User navigates to Login page.

2. User enters username and password.

3. User clicks “Log In”.

4. System verifies credentials.

5. System redirects user to appropriate dashboard.

Alternative Flow 1 (Invalid Credentials):

Occurs at Step #: 4

1. At Step 4, credentials are incorrect.

2. System displays error message.

3. User is prompted to re-enter credentials.

3. Use Case Name: Enforce Role-Based Access

Primary Actor: System

Goal:

Preconditions:

- User is logged in

Postconditions:

- Unauthorized access is denied to user

- Authorized access is available to user

Basic Flow:

1. User attempts to access a resource.

2. System checks user role.

3. If role is authorized, access is granted.

Alternative Flow 1 (Unauthorized Access):

Occurs at Step #: 3

1. At Step 3, role is not authorized.

2. System displays “Access Denied.”

3. System logs the attempt.

Use Case Name: View List of Advised Students

Primary Actor: Faculty

Goal: Faculty is able to see list of advised students

Preconditions:

- Faculty is logged in

Postconditions:

- List of advisees is displayed

Basic Flow:

1. Faculty logs in.

2. Faculty selects “My Advisees.”

3. System retrieves assigned students.

4. System displays student list.

Alternative Flow 1 (No Assigned Students):

Occurs at Step #: 3

1. At Step 3, no students found.

2. System displays “No assigned advisees.”

Use Case Name: View Student Transcript (Faculty)

Primary Actor: Faculty

Goal: Faculty can view student transcripts

Preconditions:

- Faculty is logged in

- Student is assigned advisee

Postconditions:

- Transcript is displayed

Basic Flow:

1. Faculty selects a student.

2. Faculty clicks “View Transcript.”

3. System retrieves transcript data.

4. System displays courses, grades, and GPA.

Alternative Flow 1 (Student Not Assigned):

Occurs at Step #: 3

1. At Step 3, system detects unauthorized access.

2. System denies request and displays error.

Use Case Name: View Student Transcript (Student)

Primary Actor: Student

Goal: Student is able to view their own transcript

Preconditions:

- Student is logged in

Postconditions:

- Student transcript displayed

Basic Flow:

1. Student clicks “My Transcript.”

2. System retrieves transcript.

3. System displays transcript.

Alternative Flow 1 (Transcript Unavailable):

Occurs at Step #: 2

1. At Step 2, transcript not found.

2. System displays error message.

Use Case Name: View Degree Plan

Primary Actor: Student or Faculty

Goal: Students and faculty can view the degree plan

Preconditions:

- User is logged in

Postconditions:

- Degree plan is displayed

Basic Flow:

1. User selects “Degree Plan.”

2. System retrieves required courses.

3. System displays completed and remaining courses.

Alternative Flow 1 (Degree Plan Not Found):

Occurs at Step #: 2

1. At Step 2, system cannot locate degree plan.

2. System displays error message.

Use Case Name: View Prerequisites of Courses

Primary Actor: Faculty or Student

Goal: Faculty and students can view the prerequisites of courses

Preconditions:

- Faculty is viewing a course that requires prerequisites

- Student is viewing a course that requires prerequisites

Postconditions:

- Prerequisites of a course will be displayed when course is selected

Basic Flow:

1. User selects a course.

2. System identifies that this course has prerequisite(s).

3. System retrieves prerequisite data.

4. System displays course with prerequisites.

Alternative Flow 1 (Missing Prerequisite Data):

Occurs at Step #: 3

1. At Step 3, prerequisite data unavailable.

2. System displays available course info only.

Alternative Flow 2 (Course Has No Prerequisites):

Occurs at Step #: 2

1. At Step 2, system won't identify any prerequisite data.

2. Courses with no prerequisites will display "No prerequisites".

9. Use Case: Generate Course Recommendation
   Primary actor: Student or Faculty

   Goal: User is able to have a generated course recommendation.

   Preconditions:
   - User is logged in
   - Transcript and degree plan exist

   Post Conditions:
   - Recommended courses displayed

   Basic Flow:

   1.User clicks "Generate Recommendation."

   2. System looks at degree map.

   3. System analyzes completed courses.

   4. System checks prerequisites.

   5. System generates recommended schedule.

6. System displays recommendation.

Alternative Flow 1 (No Eligible Courses):

Occurs at Step #: 5

1. At Step 5, no eligible courses found.
2. System displays message: “No remaining required courses.”

Use Case Name: Advising Report

Primary Actor: Faculty

Goal: Generate a report containing completed coursework and recommended future courses for advised student.

Preconditions:

Faculty member is logged in
Student is selected from advisee list.
Student academic data exists in the system.

Post Conditions:

Report is generated.
Faculty is able to best advise a student based off report.

Basic Flow:

Faculty selects a student from the advisee list.
Faculty clicks “Generate Advising Report.”
System retrieves the student’s transcript and degree plan.
System analyses completed coursework and missing requirements.
System generates a report including completed courses and recommended future courses.
System displays advising report.

Alternate Flow 1 (Missing Academic Data):

Occurs at Step #: 3

At Step 3, student data is missing or incomplete.
The system displays a warning message.

System generates a partial report, if possible.

Alternate Flow 2 (No Degree Plan):

Occurs at Step #: 3

At Step 3, student's degree plan isn't available or doesn't exist.
The system displays a warning message.
System doesn't generate anything because it doesn't have degree information.

3. Peer Evaluations:

Team Members:	Completion of the assigned task (0-20)	Team Engagement (0-5)	Justifications for point deductions
Isabella Morgan	20	5	N/A
Ingrid Alvarez	20	5	N/A